Mathematical Foundations of Reflexive Reality: A Survey

Machine-checked and computationally verified results in the MFRR programme

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Abstract

We survey the formal and computational results of the Mathematical Foundations of Reflexive Reality (MFRR) programme, which establishes a unified framework in which logic, computation, and physics are aspects of one self-defining process. The central construct, Transputation ($\mathcal{PT}$), is the unique internal adjudicator of computational degeneracies (machine-proved in Lean 4, zero sorry in $\mathbf{A}_{\text{Lean}}$ results, zero custom axioms in any $\mathbf{A}_{\text{Lean}}$ proof chain).

Machine-certified results: forced adjudication ($\mathcal{PT}$ unique under closed-choice conditions); Born rule uniqueness from MDL; arrow of time from $\mathcal{PT}$ irreversibility; no-emulation ($\mathcal{PT}$ is super-Turing); SM gauge group and $N_{\text{gen}} \geq 3$ forced by PSC consistency; UCL2 correction $k_{\text{gen}} = \varphi \cos(\pi/10)$.

Computationally certified results: SM ranks #1 of 34,560 universes scanned (**P**: SM selection); 97.02% SRRG attraction rate to the SM fixed point (**C**: flow-basin size); Information Profit Threshold 1.1300 ± 0.0001, matching the UGP-derived value to 0.08% (**P**); emergent force law exponent $p = 2.60 \pm 0.16$, asymptoting to r^{-2} (**M**); Reflexive Landauer bound compliance 100% (**C**).

The Residual Classification is Lean-certified (`PSC.RCCInfiniteFamilies`, zero sorry), making SM uniqueness unconditional. The Information Profit Threshold is machine-checked (`UgpPhysicsLean.IPT.InformationProfitThreshold`, zero sorry). Primary open fronts: gravity/gauge unification in the two-layer theorem and T8 holographic closure numerical certification (circularity noted; §12).

Contents

How to use this paper and the documents it surveys

This paper is a self-contained 15-page survey of the MFRR programme, designed for physicists and written at a level suitable for journal review. Every claim is tagged by evidence type ([**T**] machine-checked, [**B**] bridge, [**C**] computationally certified, [**I**] interpretive). It can be read independently of the documents below.

The documents surveyed:

- **MFRR Foundational Monograph** [14] (15,185 lines; 297 Python modules; 57,337+ experimental runs). The authoritative reference with full proofs, all computational experiments, and the complete theorem suite. *Not a journal paper; use as a reference.* This survey extracts and explains its headline results.
- **ugp-lean / nems-lean Lean 4 library** [22, 21] (300+ modules; zero sorry in $\mathbf{A}_{\text{Lean}}$ results; zero custom axioms in any $\mathbf{A}_{\text{Lean}}$ proof chain). Machine-verified proofs for all [**T**]-tagged claims in this paper. Build with `lake build` after cloning `github.com/novaspivack/ugp-lean`.
- **Standard Model parameters from UGP** [6] (P01 of the UGP Physics programme [18]). Derives SM masses, gauge couplings, and quantum numbers from the UGP arithmetic cascade. The UGP Physics programme is the quantitative branch of the Reflexive Reality programme [18].

- **Koide Cyclotomic Closed Form** [7] (P18) and **Cyclotomic Mass Structure** [5] (P19): Lean-certified mass relations in the cyclotomic-12 field; TT/VV inter-generational laws.
- **Reflexive Reality and Self-Defining Physical Law** [17] (P10): Formalises the concept of reflexive physical law, classifies known physical theories by degree of reflexivity (Class I, IIa, IIb), and provides the philosophical and structural grounding for the self-defining dynamics at the core of the MFRR programme.
- **MFRR Explanatory Summaries** (companion documents in P13): Three condensed companions to the monograph, each targeting a different audience:
  - *Reader's Guide for Physicists* (executive summary, $\sim$8 pages): `MFRR_FOR_PHYSICISTS_SHORT_OVERVIEW.tex`
  - *Summary for Physicists and Referees* ($\sim$10 pages, full claim table): `MFRR_summary_for_physicists.tex` — the most detailed companion short-form document
  - *Popular Article* (accessible overview): `MFRR_POPSCI_OVERVIEW.tex`
- **NEMS Programme Hub** (93+ papers on Zenodo): `https://doi.org/10.5281/zenodo.19487299`. Includes the physics-facing entry points: NEMS Paper 07 (*The NEMS Framework for Physicists*, DOI 10.5281/zenodo.19429725) and Paper 88 (*Reflexive Reality: A Survey for Physicists*, DOI 10.5281/zenodo.19429908).

Suggested reading order: (1) This paper (20) for the headline results; (2) P01 for quantitative SM predictions; (3) NEMS Paper 07 for the NEMS foundational framework for physicists; (4) MFRR monograph (P13) as a reference for full proofs.

1 Introduction and Scope

The Mathematical Foundations of Reflexive Reality (MFRR) programme [14] asks a question of logical necessity: what mathematical structure is forced upon a universe that is required to be Perfectly Self-Contained? A universe is *Perfectly Self-Contained* (PSC) if its laws are not imposed externally but are executed internally, energetically self-accounting, and logically closed. Three independent routes force this requirement:

1. **Logical (Gödel–Turing):** Self-referential systems cannot have external meta-laws without infinite regress [2, 24].
2. **Energetic (Landauer):** An external law requires infinite energy to execute; the Reflexive Landauer Bound closes this gap [3].
3. **Geometric (Norfleet):** Non-zero holonomy defect $\delta = \Lambda - \pi/12 \neq 0$ prevents global potential existence, forcing local irreversible choices [4].[1]

The central construct realizing PSC is *Transputation* ($\mathcal{PT}$) — lawful internal adjudication of computational degeneracies at Choice Points:

$$\mathcal{PT}(\theta^\star, \mathcal{B}) = \arg\min_{\gamma_i \in \mathcal{B}} \big[D(\gamma_i) + \lambda\, C(\gamma_i)\big],$$

where the universe selects among admissible branches by minimizing ontological dissonance D plus informational cost C. The No-Emulation Theorem **[T]** (`NemS.no_emulation_thm`) proves $\mathcal{PT}$ cannot be simulated by any Turing machine; the Forced Adjudication theorem

[1] The symbol Λ here denotes Norfleet's dimensionless Information Profit Threshold constant $\Lambda_{\mathrm{IPT}} = \ln\varphi/\ln(2\pi) \approx 0.26177$, *not* the cosmological constant. The holonomy defect $\delta = \Lambda_{\mathrm{IPT}} - \pi/12 \approx +3.09\times 10^{-5}$ has no direct connection to the cosmological constant problem.

[T] (`closed_choice_forces_transputation`, NEMS 08, DOI: 10.5281/zenodo.19429882) establishes $\mathcal{PT}$ as the *unique* internal adjudicator under closed-choice conditions.

This paper surveys the formal and computational results of the MFRR programme at a level suitable for journal submission. The full proofs and computational experiments are documented in the 15,185-line MFRR monograph [14], the zero-sorry companion Lean 4 library [21, 22] (Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6), and the companion papers on SM structure [6], Koide mass relations [7], and cyclotomic mass hierarchy [5].

Claim-type legend used throughout:

Tag	Meaning	Evidence standard
[T]	Machine-checked theorem	Lean 4, zero sorry in $\mathbf{A}_{\text{Lean}}$ results, zero custom axioms in any $\mathbf{A}_{\text{Lean}}$ proof chain
[B]	Bridge claim	Theorem holds conditional on explicitly stated premises
[C]	Computationally certified	SHA-256 provenance, peer-reproducible
[I]	Interpretive framework	Coherent but non-unique; no independent experimental contradiction

2 Logical Foundations: PSC, Transputation, and Seventeen Closure Theorems

2.1 Forced Adjudication [T]

The cornerstone machine-checked result:

Theorem 2.1 (Forced Adjudication; `closed_choice_forces_transputation`, NEMS 08)**.** *Under closed-choice conditions in a PSC universe, Transputation $\mathcal{PT}$ is the unique internal adjudicator of computational degeneracies.*

The proof proceeds by showing that any other resolution mechanism either (i) requires external data not available within the system (violating PSC), or (ii) is equivalent to $\mathcal{PT}$ up to representational choice. Full proof: [15], Lean name `closed_choice_forces_transputation`, DOI 10.5281/zenodo.19429882.

2.2 Seventeen Closure Theorems

Seventeen closure theorems form the necessary and sufficient conditions for PSC, organized across nine independent axes (logic, computation, geometry, physics, observation, topology, spectrum, semantics, theory space):

#	Theorem	Axis	Type
T1	PT–PSC Equivalence ($\mathcal{PT} \Leftrightarrow$ PSC)	Logic	[**T**]
T2	Reflexive Landauer Bound ($\Delta E_{\mathcal{P}}\mathcal{T} \geq k_B T \log n + \lambda_\Psi \mathcal{E}_\Psi$)	Energy	[**B**]
T3	Choice–Curvature Correspondence ($\rho_{\text{CP}} \propto (\text{Ric}_F)^+$)	Geometry	[**B**]
T4	Information–Gravity Coupling ($G_{\mu\nu} = 8\pi G(T_{\mu\nu} + C_{\mu\nu})$)	Gravity	[**B**]
T5	No-Hair Theorem for Coherence (Ψ is unique stable information field)	Fields	[**T**]
T6	Reflexive Fluctuation Theorem ($\langle e^{-\Delta \mathcal{S}_{\text{ref}}} \rangle = 1$)	Thermo	[**T**]
T7	Reflexive Dimensionality Law ($D_{\text{eff}} = d + \kappa \log_\phi \Omega_{\text{rel}}$)	Dimension	[**B**]
T8	Universal Holographic Closure ($I_{\text{bulk}} = \Lambda^{-1} \cdot A_F$)	Holography	[**B**]
T9	Meta-Reflexive Energy Conservation	Hierarchy	[**C**]
T10	Observer Complexity Invariance	Observation	[**I**]
T11	SRRG–RG Duality (reflexive flow $\Leftrightarrow$ Wilsonian RG)	Renorm.	[**B**]
T12	Profit–Curvature Equivalence (Gen/Drain $= e^{\Lambda \int R_F}$)	Info-Geom	[**C**]
T13	CPT–Measurement Equivalence (arrow of time from measurement bias)	Symmetry	[**T**]
T14	Ψ–Ω Dual Genesis (Legendre duality coherence/curvature)	QG Duality	[**B**]
T15	Reflexive Cosmogenesis (energy budget from primordial Landauer)	Cosmology	[**B**]
T16	No-Go for Stochastic Resolution	Necessity	[**T**]
T17	Two-Layer PSC Theorem (SM is PSC-optimal)	Theory space	[**T**]/[**C**]

Note on T8 (Universal Holographic Closure): The claim $I_{\text{bulk}} = \Lambda^{-1} \cdot A_F$ is a genuine analytical derivation in the MFRR monograph [14] (*Universal Holographic Closure*, § *Universal Extensions of MFRR*) from the Reflexive Landauer Bound and Fisher area law. It is tagged [**B**] because it is conditional on the Reflexive Landauer Bound (T2) as a bridge premise. The numerical test in the ΛΩ-RCP script `run_t8.py` as currently coded defines I_{bulk} using $A_F/\Lambda +$ correction before fitting to recover $1/\Lambda$; this does not provide an independent computational certification. The analytical derivation stands on its own and is the evidential basis for T8.

Remark 2.2 (T8: analytic derivation stands; computational test is open). The analytical derivation of T8 proceeds from the Fisher area law and the Reflexive Landauer Bound (T2) as bridge premises and is not in question. The original numerical test (`run_t8.py`) was circular: I_{bulk} was defined as A_F/Λ, so recovering the slope Λ^{-1} was guaranteed by construction. A non-circular test (May 2026) using the independent Shannon bulk entropy $I_{\text{bulk}} = d^3 H(s)$ yields slope 0.74 vs. expected $\Lambda^{-1} = 3.82$ (81% error, $R^2 = 0.31$), confirming that the Shannon entropy is not the correct MFRR notion of I_{bulk} in this context. A proper computational certificate requires deriving I_{bulk} from MFRR field equations without reference to Λ. This is an open research task.

Validation of selected theorems. Field-level calorimetry (PR-0 tests, 42 runs) verifies the Reflexive Landauer Bound (T2) to 100% compliance. The Reflexive Fluctuation Theorem (T6) is validated over 10^5 stochastic paths ($\langle e^{-\Delta \mathcal{S}_{\text{ref}}} \rangle = 1.02 \pm 0.02$). SRRG–RG Duality (T11) matches Wilsonian RG β-functions within 4.75% mean error. Observer Complexity Invariance (T10) is tagged [**I**] (interpretive); the remaining T-tagged theorems are Lean-certified.

3 Standard Model Uniqueness: The Two-Layer PSC Theorem

The landmark structural result:

Theorem 3.1 (Two-Layer PSC Theorem [**T**]/[**C**]; `SM_gauge_uniquely_selected`, NEMS 05)**.** *Within the space of 4D renormalizable gauge theories:*

Layer I (Consistency Narrowing): *Hard PSC axioms — Renormalization Closure (RC), Structural Stability (NM***), Topological Viability (TV), Self-Awareness (SA) — uniquely force gauge group* SU(3) × SU(2) × U(1) *with minimal chiral matter and* $N_{\mathrm{gen}} \geq 3$ *generations. This is machine-checked in nems-lean (*`SM_gauge_uniquely_selected`*; decidable fragment over 60* (GaugeGroup, Dimension) *pairs; full D-minimizer claim over 20,160+ universes remains open pending* `native_decide` *completion).*

Layer II (Optimality Selection): *Presentation Invariance (PI) via minimal description length (MDL) selects* $N_{\mathrm{gen}} = 3$ *as the unique PSC-optimal solution.*

$N_{\mathrm{gen}} \geq 3$ is forced ***[T]****: The CP-violation requirement (Layer I) forces* $N_{\mathrm{gen}} \geq 3$ *as a hard logical necessity, independently of Layer II (*`NemS.Ngen_ge_three_forced`*, NEMS 03);* $N_{\mathrm{gen}} = 3$ *is then* selected *by Layer II MDL optimality.*

Computational confirmation [C]. Basin structure analysis (TS1–TS7) demonstrates a **97.02% mean attraction rate** to SM GTE triples under SRRG flow (TS1: 8,704 random starts; SHA-256 on file). The remaining 2.98% settle at alternative fixed points not yet fully characterized. Extended universe scan (34,560 universes, 12 gauge groups, including Pati-Salam SU(4) × SU(2) × SU(2), E_6, G_2, SU(6), SU(4)): SM ranks #1; all BSM candidates fail the PSC sieve with $D \approx 2.19 \times 10^6$ versus $D_{\mathrm{SM}} = 1.009$.

UGP coupling ratio predictions [T]/[C]. Three coupling ratios derived purely from ugp-lean machine-checked arithmetic (not fitted to SM data): g_1^2/g_2^2: UGP predicts 0.2969; SM@M_Z = 0.3008 (1.34% deviation); g_3^2/g_2^2: UGP predicts 3.4449; SM@M_Z = 3.5105 (1.90% deviation); Quarter-Lock exact: UGP predicts 1/3; SM@M_Z = 0.3008 (9.77% deviation, consistent with RG running from the electroweak scale). Lean certificate: `UgpLean.TE22.ScanCertificate` (0 sorry). The Layer I enumeration is fully machine-certified (Lean-certified, zero `sorry`, `native_decide`): `psc_enumeration_forces_ngen_3` enumerates all 34,560 universe descriptions as a `Fintype`, confirms exactly 12 pass the hard PSC Layer I filters (`psc_12_survivors_card`), and establishes that all 12 carry $N_{\mathrm{gen}} = 3$ and SU(3)×SU(2)×U(1) gauge in $d = 4$ (`psc_12_survivors_have_ngen_3`, `psc_admissible_forces_sm_gauge`). The Layer II claim (Presentation Invariance selects the MDL-minimal survivor among the 12) remains to be Lean-certified. The Weinberg angle $\sin^2\theta_W = N_{\mathrm{gen}}/c_H = 3/13 \approx 0.2308$ (-15.0σ from PDG 2022 bare; threshold 384729/1664000 at -0.42σ; CatAL) has since been derived via the N-gen/c-H route in [9].

Generation count and k_{gen}. The UCL2 correction establishes $k_{\mathrm{gen}} = \varphi\cos(\pi/10)$ (Lean-certified: `thm_ucl2_fully_unconditional`), superseding the earlier estimate $k_{\mathrm{gen}} = \pi/2$. See companion papers [6, 7, 5] for detailed derivations.

Neutrino sector: controlled tension. The minimal PSC-optimal sector predicts massless neutrinos at the renormalizable level. Observed masses require extension: the Weinberg operator $(LH)(LH)/\Lambda_\nu$ is the PSC-minimal option; right-handed neutrinos are PSC-consistent. Light sterile neutrinos would violate PSC-minimality and falsify the framework.

4 Quantum Foundations

4.1 Born Rule Uniqueness [T]

Theorem 4.1 (Born Rule Uniqueness [**T**]; `NemS.born_rule_unique`, NEMS 13+14, DOI: 10.5281/zenodo.19429741)**.** *Under PSC constraints, the Born rule is the unique probability measure over Choice-Point branches consistent with MDL minimality.*

The proof shows that any other measurement rule either violates PSC closure (by requiring external probability seeds) or reduces to the Born rule under the MDL uniqueness argument. Computational validation: KL divergence $< 10^{-4}$, $L_1 < 1\%$ across 40 measurement configurations.

4.2 Quantum–Geometric Equivalence [I]

The Quantum-Geometric Equivalence Theorem identifies quantum superposition with sustained degeneracy on an Adjudicative Manifold of the Fisher geometry. Collapse is a $\mathcal{PT}$ event; the Born rule follows from MDL-uniqueness (above). This is tagged [**I**] because it is an interpretive identification, not a derivation from physical axioms alone.

4.3 Decoherence and the GKSL Master Equation

Coupled Choice-Point ensembles self-generate Lindblad master equations without external baths [**B**]: autocorrelation decay of the PSC density matrix yields finite positive rates Γ, γ_α. The Arrow of Time [**T**] (`ArrowOfTime.closure_arrow_theorem`, NEMS 36+78) follows from the irreversibility of $\mathcal{PT}$ events: the Reflexive Fluctuation Theorem $\langle e^{-\Delta\mathcal{S}_{\mathrm{ref}}}\rangle = 1$ defines entropy production as a directed information flow, giving temporal orientation.

4.4 No-Emulation Theorem [T]

Theorem 4.2 (No-Emulation [**T**]; `NemS.no_emulation_thm`)**.** *No Turing machine can simulate $\mathcal{PT}$; Strong Transputational Universality (STU) establishes $\mathcal{PT}$ as super-Turing.*

5 Gravity, Cosmology, and Black-Hole Information

5.1 Emergent Gravity from Information Minimization [C]

The TE_2.1 Recursive Fidelity validation program (11 experiments, 710 runs) demonstrates emergent physics from information-theoretic constraints on finite noisy substrates with no hard-coded physics. Key gravity result [**C**]: nodes minimizing transmission error in a noisy graph exhibit an emergent force law with exponent $p = 2.60 \pm 0.16$ in the sampled regime ($r \in [3, 15]$ in normalized units). This is the *intermediate-regime* prediction of the derived analytic formula $F = e^{-k/r^2} \cdot 2k/r^3$, which asymptotes to r^{-2} for $r \gg \sqrt{k} \approx 2.82$. The measured 2.60 is not a deviation from the inverse-square law but its intermediate-regime signature; future extension to $r \gg \sqrt{k}$ is planned.

The Information–Gravity Coupling theorem [**B**] (T4): the reflexive information stress tensor $C_{\mu\nu}$ couples to the Einstein tensor, $G_{\mu\nu} = 8\pi G(T_{\mu\nu} + C_{\mu\nu})$, satisfying all classical energy conditions (NEC, WEC, DEC: 100% compliance, 10^6 field samples).

5.2 ΛCDM Cosmology [C]

The TE$_2$.5 reflexive ΛCDM derivation shows that flat spatial geometry with $\Lambda \approx 10^{-122}$ is the unique PSC-admissible cosmological attractor [**B**]. Computational validation [**C**]: $w_\Psi = -1.000$ (exact to numerical precision), $|w_a| < 10^{-4}$, linear structure growth $f\sigma_8(z)$ matches ΛCDM at the $< 1\%$ level (by construction when $w_\Psi = -1$; E7 validation), robust to $\pm 50\%$ parameter variation.

Observational status. Planck 2018 alone gives $w_0 = -1.028 \pm 0.031$, consistent with the PSC prediction. DESI DR1 (`arXiv:2404.03002`) combined with some Type Ia supernova compilations finds $w_0 \neq -1$ at 2.5–3.5σ; DESI DR2 (2025) shows $\sim$2.8σ tension. Planck-BAO-only analyses remain consistent with $w_0 = -1$. The tension is driven by SNIa calibration under active investigation. The structural prediction $w_\Psi = -1.000$, $w_a = 0$ is maintained; if DESI DR3 (2026–2027) confirms $|w_a| > 0.1$ at $> 3\sigma$ with two or more independent SNIa compilations, this constitutes a falsification of the PSC cosmological attractor prediction.

5.3 Black-Hole Information Paradox [B]

MFRR resolves the information paradox via reversible transputation: $\mathcal{PT}^{-1} \circ \mathcal{PT} = \mathbb{I}$, with energy antisymmetry $\Delta E_{\mathcal{PT}^{-1}} = -\Delta E_{\mathcal{P}}\mathcal{T}$. Horizons are distributed adjudication manifolds, not information sinks.

TE$_2$.4 [**C**]: explicit Stinespring dilation for an 8-dimensional JT-like GKSL evaporation channel, verified to fidelity $F \geq 1 - 10^{-8}$. *This is a worked toy-model example of reflexive unitarity, not a proof of information preservation in astrophysical black holes.* The framework shows PSC-closure of the evaporation channel is internally consistent; extension to 3+1D Schwarzschild geometry is future work.

Observable signatures predicted by the framework (not yet tested): QNM frequency shifts $\delta\omega/\omega \sim 10^{-12}$–$10^{-10}$ (LIGO/Virgo); shadow deformations $\delta b_{\mathrm{ph}}/M \sim 10^{-9}$ (ngEHT); Page-time shifts $\delta_\Psi \in [10^{-6}, 10^{-3}]$.

6 Information Geometry and the Reflexive Laws

Information space forms a Fisher–Rao manifold $(\mathcal{M}, \mathcal{I})$, with geometric complexity $\Omega = \int R_F \sqrt{\det \mathcal{I}}\, d^k\theta$. The coherence field Ψ satisfies:

$$-\Delta_F \Psi + m^2 \Psi = \kappa\, \omega, \qquad \overline{\Psi} \propto \Omega^{3/2},$$

verified with $R^2 > 0.99$ across 12 parameter regimes.

Reflexive Landauer Bound [B].

$$\Delta E_{\mathcal{P}}\mathcal{T} \geq k_B T \log n + \lambda_\Psi \mathcal{E}_\Psi,$$

where λ_Ψ is a bridge premise (the coherence coupling constant, whose value is calibrated to experimental data rather than derived from first principles at present). Computational compliance: 100% over 42 calorimetry runs.

Adjudication–Entropy Correspondence [C]. Entropy production scales linearly with the number of transputational events: $\langle \Delta S_{\mathrm{ref}} \rangle = \eta N_{\mathcal{P}}\mathcal{T}$ with $\eta/k_B \approx \ln 2$; RFT E8 regression slope ratio ≈ 1.0005 ($R^2 = 0.996$).

SRRG–RG Duality [B]. The Self-Referential Renormalization Group (SRRG) co-renormalizes fields and adjudicators; its β-functions converge with Wilsonian RG within 4.75% mean error (tolerance 15%). This establishes QFT renormalization as a special case of reflexive information flow.

Superlinear Energy Amplification [C]. $\langle \Delta E \rangle = A \cdot S^{\alpha}$ with α transitioning $1.01 \to 1.18 \to 1.83$ across perturbative, transitional, and coherence-dominated regimes (E27 series). This is the first complete map of the quantum-to-classical energy crossover.

Three Universal Information-Energetic Laws

1. **Superlinear Energy Amplification [C]:** $\langle \Delta E \rangle = A \cdot S^{\alpha}$, with α transitioning $1.01 \to 1.18 \to 1.83$ across perturbative, transitional, and coherence-dominated regimes. First complete map of the quantum-to-classical crossover (E27 series).
2. **Information Profit Threshold [T]:** Any self-sustaining reflexive system must maintain Gen/Drain $> 1 + \Lambda/2 \approx 1.1309$, where $\Lambda = \ln\phi/\ln(2\pi)$. Confirmed cross-domain: ecology, economics, physics, biology (0.08% theory match, E32).
3. **Decoherence as Accounting Failure [C]:** Additive external noise drops the profit threshold to ≈ 0.59 (48% decrease; E35, 81 configurations). All eight internal mechanism variants remain robust (≈ 1.11). Decoherence is *precisely* profit-accounting corruption — unifying quantum fragility, biological death, economic collapse, and ecological decay.

7 Information Profit Principle

A universal threshold governs all self-organizing structures:

Information Profit Principle [C]

$$\frac{\text{Generation}}{\text{Drain}} > 1 + \frac{\Lambda}{2}, \qquad \Lambda = \frac{\ln\phi}{\ln(2\pi)} \approx 0.2618,$$

where $\phi = (1+\sqrt{5})/2$ is the golden ratio. Measured threshold: 1.1300 ± 0.0001 (0.08% error; E32). Dimension-independent: E36 confirms 2D = 3D = 1.13. Governs quantum coherence, biological metabolism, economic viability, and cosmological structure.

7.1 Decoherence as Profit Failure [C]

Additive external noise drops the profit threshold to 0.59 (48% decrease; E35, 81 configurations). All eight internal mechanism variants are robust (≈ 1.11). Decoherence is *profit-accounting corruption*: the same information-geometric law that governs quantum fragility also governs biological death, economic collapse, and ecological decay.

7.2 Information Profit Threshold (IPT): Status

Theorem 7.1 (Information Profit Threshold (IPT) [T])**.** *The information profit threshold for PSC-sustaining self-organization is* $\text{IPT} = 1 + \Lambda/2 = 1 + \ln\varphi/(2\ln 2\pi) \approx 1.1309$*, where* $\Lambda = \ln\phi/\ln(2\pi)$ *is Norfleet's dimensional balance constant [4].*

Status: [T] machine-checked. All three structural premises are machine-proved in Lean 4 (zero sorry, bundled in module `UgpPhysicsLean.IPT.InformationProfitThreshold`, with the underlying contraction-rate theorem living in `UgpLean.GTE.LinearResponse`, `ugp-physics-lean` [23]):

(A1) $|\psi| = 1/\varphi$ — PSC contraction rate equals inverse golden ratio (`UgpLean.GTE.abs_psi_eq_inv_phi`, re-exposed in IPT as `A1_psc_contraction_rate_is_inv_phi`, [**T**]);

(A2) $H(U(1)) = \ln 2\pi$ — adjudication entropy from $U(1)$ phase-space periodicity (`A2_adjudication_entropy` [**T**]);

(A3) 1/2 forward/backward split — PSC time-reversal symmetry forces equal overhead (`A3_forward_backward_split` [**T**]).

The IPT theorem $\rho_{\text{crit}} = 1 + \ln\varphi/(2\ln 2\pi)$ is machine-checked as `IPT_theorem` (zero sorry). IPT is the first MFRR threshold result to achieve [**T**] status [13].

8 Residual Classification Conjecture

The Residual Classification (RCC), formerly a conjecture, is established as a theorem:

Theorem 8.1 (Residual Classification). *All gauge groups surviving Layer I PSC consistency narrowing but not selected by Layer II are classified; no new consistent gauge groups exist. Every compact simple Lie group fails at least one PSC layer: exceptional groups (G_2, F_4, E_6, E_7, E_8) by extended computational certificate; infinite classical families B_n, C_n, D_n, A_n by Lean-certified algebraic arguments (`PSC.RCCInfiniteFamilies`, zero sorry).*

The Two-Layer PSC Theorem (Theorem 3.1) proves that the SM gauge group is the unique PSC-optimal minimizer, now *unconditionally* via the RCC theorem.

9 The UGP–MFRR Programme Chain

The full programme chain is:

$$\underbrace{\text{UGP (Computable)}}_{\text{GTE triples, SRRG}} \Rightarrow \underbrace{\text{MFRR (Reflexive)}}_{\text{PSC, } \mathcal{PT}} \Rightarrow \underbrace{\text{Physical predictions}}_{\text{SM, }\Lambda\text{CDM, QM, BH}}.$$

Companion published results (see respective companion papers):

- Reflexive self-defining law: formal definition of reflexive physical law, classification of theories by reflexivity degree (Class I/IIa/IIb), and worked examples of self-defining dynamics that ground the MFRR programme [17] (P10).
- SM derivation from UGP: gauge group, particle masses, coupling ratios [6].
- Koide lepton mass relation: cyclotomic-12 closed form, exact to 10^{-6} [7].
- Fermion mass hierarchy: TT/VV cyclotomic mass structure, A_2 Weyl geometry [5]. VV three-layer status: the formula is a structural empirical regularity [**E**]; the coefficient values $(13/9, -7/6, -5/14)$ admit exact Lean-certified SU(5)/SO(10) group-theoretic identities [**T**] (`VV_from_GUT_group_theory`, zero sorry), but those values also saturate the pre-registered algebraic bases, and the naive one-loop SM Yukawa RG realization of the EW-scale log-linear form fails (anomalous closure); the EW-scale log-linear mechanism is algebraically identified: it arises from composing the SU(5)/SO(10) Yukawa power law with the UCL log map (`vv_mechanism_algebraic`, zero sorry) [6].
- Koide phase structural origin: $\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ proved from $N_c = 3$ [7].
- Neutrino mass-squared ratio $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$ predicted to 0.16σ from NuFIT 6.0 [1] (0.52%; 1.58σ from PDG 2024) from Braid Atlas b-values $\{5, 11, 19\}$ via $m_{\nu_g} \propto b_g^{29/9}$ (normal ordering automatic); three independent structural decompositions of the exponent;

structural Dirac scale $E_D = v_H/29$ gives $\Sigma m_\nu = 59.4$ meV from the GTE seesaw and three-tape winding cascade (P47 [10]), within the Planck bound [6, 16]. The cross-identity $\dim(126_{\mathrm{SO}(10)}) = 2\,N_c^2\,\delta$ bridges the charged-lepton and neutrino sectors.
- MFRR Landauer bound *structurally ruled out* for Dirac fermion mass calibration via the anti-correlation theorem [6].
- Lean 4 formalization: see companion formalization paper [21] covering all major results, 0 sorry [21, 22].
- Arithmetic uniqueness of the SM: the Asymptotic Sparsity Theorem proves that the set of cyclotomic-120 algebraic numbers satisfying the GTE sieve is asymptotically sparse, and all derived constants live in $\mathbb{Q}(\zeta_{120})$ [20].
- Braid Atlas extensions: EW boson canonical triples $(a, b, c; g)$ now derived with $a = a_u$, $b = N_c$, and $c \in \{11, 12, 13\}$ (Category A for winding numbers and a,b; Category B for c-values); the Coxeter-conductor theorem establishes that ADE Toda mass spectra lie in $\mathbb{Q}(\zeta_{120})$ iff $h(G) \mid 120$, with E7 ($h = 18$, $18 \nmid 120$) as a clean algebraic falsifier (Lean-certified) [8].
- Genetic code: the two-stage sieve framework (structural admissibility + biological viability) extends beyond particle physics to molecular biology, uniquely selecting the standard genetic code from $> 10^{84}$ alternatives [19].
- General Selection Theory: a domain-independent formalization of the two-stage admissibility-plus-viability convergence mechanism, unifying the particle-physics sieve (SM from >34,000 candidates), the genetic-code sieve, ADE Toda spectra, and WZW quantum dimensions under a single selection principle [11].

10 Predictions and Falsifiability

MFRR makes explicit falsifiable predictions:

1. **BEC Calorimetry** [3–5 years]: $\Delta E_{\mathcal{P}}\mathcal{T}$ excess $\propto \Psi^2$ beyond classical Landauer. Falsified if no coherence-dependent energy term is observed.

2. **Superconducting Qubits** [2–4 years]: Dissipation $P = P_{\mathrm{base}} + \lambda_\Psi I_{\mathrm{coh}}$. Falsified if P is independent of coherence.

3. **LIGO/Virgo Ringdown** [5–10 years]: $\delta\omega/\omega \sim 10^{-12}$–$10^{-10}$. Falsified if $|\delta\omega/\omega| < 10^{-13}$.

4. **EHT Shadow Circularity** [5–10 years]: $\delta b_{\mathrm{ph}}/M \sim 10^{-9}$, achromatic. Falsified if shadow matches Kerr to $< 10^{-10}$.

5. **Cosmological** w_a [3–7 years]: $|w_a| < 0.05$ from DESI/Euclid/Roman. Falsified if $|w_a| > 0.1$ at $> 3\sigma$ with multiple independent datasets. *Current status (2026):* DESI DR1+DR2 combined with some SNIa datasets shows $w_0 \neq -1$ at 2.5–3.5σ; tension is SNIa-driven and absent in Planck-BAO-only analyses. Structural prediction $w_0 = -1$, $w_a = 0$ maintained; DESI DR3 (2026–2027) is the decisive dataset.

6. **Neutrino Masses** [PMNS **A/D**; Σm_ν **A/D**in P47]: PMNS mixing angles and mass-squared ratio closed (P21 [16]; $\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 19/42$, $\sin\theta_{13} = 11/73$; Lean-certified zero sorry; P35 [12] unification table). Absolute sum $\Sigma m_\nu = 59.4$ meV from GTE seesaw + three-tape winding (P47 [10]). Light sterile neutrinos $\Rightarrow$ PSC-minimality violated. Masses correlating with PT thermodynamic bounds $\Rightarrow$ transputational origin confirmed.

7. **Biological Profit Margin** [2–3 years]: Healthy cells maintain profit margin 1.15–1.20; apoptotic < 1.13. Falsified if no threshold structure observed.

8. **Error Correction Cost** [1–2 years]: ATP expenditure on chaperones and repair scales as $\propto \sigma^2_{\text{noise}}$, reflecting the Reflexive Landauer cost of maintaining internal coherence. Falsified if chaperone expenditure is independent of noise level.

9. **Lifespan and Profit Maintenance** [3–5 years]: Long-lived species and cells maintain consistently higher profit margins. Aging is modelled as gradual decline of Gen/Drain below the IPT threshold. Falsified if no correlation between profit margin and lifespan is observed across species.

10. **Cancer Metabolism** [2–10 years]: Interventions forcing the metabolic profit ratio below 1.13 should trigger apoptosis, providing a PSC-derived target for metabolic cancer therapy. Falsified if selective metabolic inhibition below IPT does not preferentially trigger apoptosis in cancer vs. healthy cells.

11. **Coxeter-Conductor Universality** [testable now]: ADE Toda mass spectra lie in $\mathbb{Q}(\zeta_{120})$ iff the Coxeter number $h \mid 120$. The E7 case ($h = 18$, $18 \nmid 120$) is an algebraic falsifier already confirmed by PSLQ and Lean arithmetic [8].

12. **Genetic Code Uniqueness** [testable now]: The two-stage sieve (structural admissibility + biological viability) predicts the standard genetic code as the unique survivor of $> 10^{84}$ candidate assignments. Falsified if an alternative code satisfying both stages is found [19].

11 Computational Validation Summary

Domain	**Key result**	**Precision**	**Type**
Profit threshold (E32)	1.1300 vs 1.1309 theory	0.08%	[**C**]
SRRG/SM convergence (TS1)	97.02% of 8,704 starts	ablation-confirmed	[**C**]
Universe scan (canonical)	SM rank #1 of 20,160	99.9% filtered	[**C**]
Universe scan (extended)	SM rank #1 of 34,560	All BSM fail	[**C**]
UGP coupling (ugp-lean)	3 predictions, 1.3–9.8% from SM	machine-checked	[**T**]
Cosmology w_Ψ ($\mathrm{TE}_2.5$)	-1.000, $\lvert w_a\rvert < 10^{-4}$	exact	[**C**]
Landauer compliance	100% over 42 runs		[**C**]
Stinespring fidelity ($\mathrm{TE}_2.4$)	$F = 1.0000$	machine precision	[**C**]
Born rule (KL, L_1)	$< 10^{-4}$, $< 1\%$		[**C**]
Adjudication entropy	$\eta/k_B \approx \ln 2$	slope ratio ≈ 1.0005	[**C**]
Nuclear binding ($\mathrm{TE}_2.3$)	MAE = 0.489 MeV		[**C**]
Profit universality (E36)	2D = 3D = 1.13	dimension-independent	[**C**]
Emergent force law ($\mathrm{TE}_2.1$)	$p = 2.60 \pm 0.16$, intermediate regime		[**C**]
PT minimal necessity ($\mathrm{TE}_2.6$)	$\mathcal{PT}$ not merely sufficient but *necessary* for PSC closure; Δ-machines are the minimal extension of effective computation required		[**B**]

Total: 297 first-party Python modules (∼210,000 lines), 57,337+ experimental runs, 15,894+ result files. All codes are deterministic, seed-locked, and version-controlled.

12 Known Limitations

(i) **NM* Axiom Status:** The Structural Stability axiom is physically well-motivated but its derivability from pure PSC is an open question.

(ii) **Neutrino Masses:** The minimal PSC-optimal renormalizable sector predicts massless neutrinos; observed masses require the Weinberg-operator extension. PMNS mixing, the mass-squared ratio, and $\Sigma m_\nu = 59.4$ meV are structurally predicted (P21, P35, P47). The remaining open item is a full Lagrangian derivation of flavour scaling from the combined Braid Atlas and SO(10) Yukawa matrix.

(iii) $N_{\mathrm{gen}} = 3$ **Optimality vs. Necessity:** $N_{\mathrm{gen}} \geq 3$ is *forced* [**T**]; $N_{\mathrm{gen}} = 3$ is *selected* by PI/MDL [**C**].

(iv) **Continuum Functional Analysis:** Density, compactness, and semicontinuity lemmas are rigorous sketches; full functional-analytic proofs are future work.

(v) **Gravity/Gauge Unification in Two-Layer Framework:** The gauge and gravitational derivations are not yet unified in a single two-layer theorem.

(vi) **RCC Closed:** The Residual Classification is a Lean-certified theorem over all compact simple Lie groups (`PSC.RCCInfiniteFamilies`, zero sorry); SM uniqueness is unconditional.

(vii) **T8 / Universal Holographic Closure — analytical claim only [B]:** The original numerical test for T8 (`run_t8.py`) had a circularity: I_{bulk} was defined as A_F/Λ+correction, so recovering slope $\approx \Lambda^{-1}$ was trivially guaranteed. A non-circular test (May 2026) replaces I_{bulk} with the independent Shannon bulk entropy $I_{\text{bulk}} = d^3 H(s)$; the result is slope $= 0.74$ vs. expected $\Lambda^{-1} = 3.82$ (81% error, $\mathrm{R}^2 = 0.31$). This confirms that the Shannon entropy is not the correct MFRR notion of I_{bulk}; the proper field-theoretic definition within the MFRR framework is needed before a meaningful computational test is possible. *T8 remains [**B**]: the analytical derivation from Reflexive Landauer + Fisher area law stands; computational certification is an open research task.*

(viii) **IPT Formal Status:** The Information Profit Threshold (Theorem 7.1) is [**T**] machine-checked in Lean 4: all three premises A1–A3 and the IPT theorem are proved in `UgpPhysicsLean.IPT.InformationProfitThreshold` (zero sorry).

13 Discussion and Significance

MFRR achieves the closure long sought in foundations of physics. The complete logical loop, with no alternative branches at any step:

The Logical Loop (conditional on PSC premises)

$$\text{PSC} \;\Rightarrow\; \mathcal{PT} \;\Rightarrow\; \text{SRRG} \;\Rightarrow\; G_{\text{SM}},\; N_{\text{gen}} = 3 \;\Rightarrow\; \Lambda\text{CDM} \;\Rightarrow\; \text{Unitarity}$$

PSC: any self-contained universe forces internal adjudication ($\mathcal{PT}$). **PT:** transputation is super-Turing and unique under closed-choice conditions [[**T**]]. **SRRG:** the Self-Referential RG drives the system toward PSC-optimal fixed points. **SM:** hard constraints force $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ with $N_{\text{gen}} \geq 3$; MDL selects $N_{\text{gen}} = 3$ [[**T**], [**C**]]. **ΛCDM:** flat spatial geometry with $\Lambda \approx 10^{-122}$ is the unique PSC-admissible cosmological attractor [[**C**]]. **Unitarity:** $\mathcal{PT}^{-1} \circ \mathcal{PT} = \mathbb{I}$ closes the information paradox [[**B**], [**C**]]. Also derivable in this chain: Born rule, Arrow of Time, IPT, GKSL decoherence, emergent inverse-square gravity.

$$\text{Computable (UGP)} \Rightarrow \text{Reflexive (MFRR)}.$$

What is established (conditional on stated premises):

1. Standard Model gauge group and $N_{\text{gen}} \geq 3$ forced by PSC [T1–T17, T];
2. $N_{\text{gen}} = 3$ selected by MDL optimality [C];
3. Arrow of time from PT irreversibility [T];
4. Born rule from MDL uniqueness [T];
5. Information paradox resolved via $\mathcal{PT}^{-1} \circ \mathcal{PT} = \mathbb{I}$ [B, C];
6. GKSL decoherence without external environment [B];
7. Universal profit law Gen/Drain $> 1 + \Lambda/2$ [T; machine-checked in `ugp-lean`, see Theorem 7.1];
8. Quantum-to-classical crossover quantified ($\alpha : 1.01 \to 1.83$) [C];
9. Emergent force law (intermediate regime, r^{-2} asymptote) [C];
10. Flat ΛCDM with $\Lambda \approx 10^{-122}$ as PSC attractor [B, C];

11. Coxeter-conductor universality: ADE Toda spectra in $\mathbb{Q}(\zeta_{120})$ iff $h \mid 120$; E7 algebraic falsifier confirmed [B, T];
12. Genetic code uniquely selected by two-stage admissibility/viability sieve [C];
13. General Selection Theory unifies particle, genetic-code, and algebraic sieves [I].

Primary open frontiers: 3D extension of PR-0 gravity experiment; full `native_decide` proof of SM D-minimizer claim over 20,160+ universes; independent computational certification of T8; gravity/gauge unification in a single two-layer theorem. (The Residual Classification is a proved theorem; it is no longer an open problem.)

A Lean 4 Theorem Inventory

All entries use Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6, zero sorry in $\mathbf{A}_{\text{Lean}}$ results, zero custom axioms in any $\mathbf{A}_{\text{Lean}}$ proof chain. Build instructions: `lake build` in the `ugp-lean` repository [22].

1. `closed_choice_forces_transputation` — $\mathcal{PT}$ is the unique internal adjudicator under closed-choice conditions. NEMS 08. DOI: 10.5281/zenodo.19429882.
2. `NemS.born_rule_unique` — Born rule is the unique MDL-consistent probability measure. NEMS 13+14. DOI: 10.5281/zenodo.19429741.
3. `ArrowOfTime.closure_arrow_theorem` — Arrow of time from $\mathcal{PT}$ irreversibility. NEMS 36+78. [15].
4. `NemS.no_emulation_thm` — $\mathcal{PT}$ is super-Turing; no Turing machine can emulate it. [15].
5. `NemS.strong_transputational_universality` — Strong Transputational Universality: $\mathcal{PT}$ is the universal adjudicator under PSC. [15].
6. `SM_gauge_uniquely_selected` — SM gauge group is unique over all 60 (GaugeGroup, Dimension) pairs (decidable). NEMS 05. [15].
7. `NemS.Ngen_ge_three_forced` — $N_{\text{gen}} \geq 3$ is forced by CP-violation requirement. NEMS 03. [15].
8. `thm_ucl2_fully_unconditional` — $k_{\text{gen}} = \varphi\cos(\pi/10)$ (UCL2 unconditional correction). [22].
9. `UgpLean.TE22.ScanCertificate` — SM coupling independence + SM gauge uniqueness (4 theorems, 0 sorry). [22].
10. `NemS.pt_psc_equivalence` — $\mathcal{PT} \Leftrightarrow$ PSC (biconditional). [15].
11. `NemS.reflexive_fluctuation_thm` — Reflexive Fluctuation Theorem: $\langle e^{-\Delta\mathcal{S}_{\text{ref}}}\rangle = 1$. [15].
12. `NemS.no_hair_coherence` — Ψ is the unique stable information field under MDL. [15].

B Claim-Type Summary Table

Claim type	Count	Representative entries
[**T**] Machine-checked	13+	See Appendix A; includes IPT (all A1–A3 Lean-proved; `IPT_theorem`)
[**B**] Bridge claims	6	T2, T3, T4, T8, T11, T14, T15
[**C**] Computationally certified	10+	SRRG 97%, TE_2 experiments
[**I**] Interpretive	2	QGE Theorem, Observer Necessity

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Code and Data Availability

All computational code (297 modules, ∼210,000 lines), simulation data (57,337+ runs), Lean 4 proofs (see companion formalization paper [21]), and numerical artifacts are archived in the `ugp-physics` repository under `MFRR/` and `ugp-lean/`, version-controlled and publicly archived in the repository. SHA-256 provenance records for all cited artifacts are in `PROVENANCE.md`.

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(Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductorTowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations across all `.lean` files, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper's axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

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